INCH-POUND

A-A-58019(USAF)
31 December 1993

Commercial Item Description (CID)
Self Generating Nitrogen Servicing Cart (SGNSC).
CID Contains Appendix A, B & C

ABSTRACT.

This CID covers a SGNSC which can be towed over unimproved surfaces and transported by helicopter and fixed wing aircraft. It will be used to produce 95% pure nitrogen gas at 5000 pounds per square inch gage (psig) at a rate of 1511 standard cubic feet per minute (scfm). The nitrogen gas will be used to service tires, struts, accumulators, and to fill external receivers. A bid sample test is required of the offeror. This CID contains Appendix A, Bid Sample Test Requirements; Appendix B, Bid Sample Test Report Requirements; and Appendix C, Certificate Format.

SALIENT CHARACTERISTICS.

1. Construction. The SGNSC shall withstand the stress, strain, shock, and vibration incident to operation, shipment, towing, sling loading under a helicopter and lifting by forklift. It shall have asbestos-free drum type brakes activated by a lever. It shall be a self-contained, enclosed, four wheel, towable cart which shall not exceed the following size and weight:

a. Size: 90 inches (in) long (tow bar up/stowed), 66 (in) wide, and 62 (in) high.
b. Weight: 4000 pounds (lb) with fuel, coolant, and oil at fill levels.

2. Electrical. The SGNSC shall use the best commercial practice electrical/elelectronic materials, components, and safety codes.

3. Ground mobility.

a. The SGNSC shall meet the mobility requirements of MIL-M-8090, Type H, Class II, Group C.
b. The SGNSC shall have the capability for being lifted by standard fork lift equipment.

Beneficial comments, recommendations, additions, clarification, and/or any other data which may improve this document, should be sent to SA-ALC/TILDD; 485 Quentin Roosevelt; Kelly AFB, TX 78241-6425.

AMSC N/A FSC 3655

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4. Air transportability. The SGNSC shall be transportable with one half fuel load and without the need for adapters or spreaders by both helicopter and fixed wing aircraft, such as the C-130, KC-135, C-17, CH-47, C-141, C-5 and C-17.

5. Maintainability.

a. The SGNSC shall require the minimum of tools for maintenance by reducing the variety of sizes and number of bolt heads and other removable fasteners.

b. The SGNSC shall not require peculiar or special tools.

c. The SGNSC shall be field maintainable. All SGNSC components, such as the membrane, engine, compressor, receivers, valves, controls, pipes, etc., shall be removable by field level maintenance technicians. Overhaul of the engine and the compressor shall be considered a depot level function.

d. Operational inspections (i.e fuel, oil, etc.) shall be performed without undue maintenance effort.

e. The SGNSC shall have an operator panel. This panel shall have the engine start control; engine stop control; nitrogen system engage/disengage control; receiver pressure gages; high and low regulator pressure gages; high and low pressure regulators; membrane temperature gage; emergency stop control; receiver isolation controls; electric glow plug control and indicator (if needed).

f. The SGNSC shall have a maintenance panel. This panel shall have a receiver drain control; system drain control; engine oil pressure gage; engine temperature gage; compressor oil pressure gage (if needed): compressor stage pressure gage; battery charge indicator; engine elapsed time indicator; compressor elapsed time indicator; a metered port and shutoff controls for connecting an oxygen gas analyzer; and tachometer.

6. The performance capability of the SGNSC shall not be affected by exposure/storage under the following environments.

a. Sand and/or dust particles as may be encountered in normal and desert operations.

b. Salt-laden moisture such as may be encountered near bodies of salt water and/or during shipboard transportation.

c. Storage at temperatures from -65 F. to +165 F

d. Fungus as may be encountered in tropical and subtropical climates.

e. Solar radiation under ambient conditions.

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f. Winds up to 45 miles per hour

g. Rain, including blowing rain.

7. When the bid sample is tested in accordance with Appendix A, the SGNSC shall meet the following nitrogen gas production and quality requirements at altitudes up to 5000 feet:

a. Production requirements:

(1) Pressure: 5000 psig.

(2) Nitrogen gas production rate: 15 ± 1 scfm.

(3) Stored nitrogen: 200 standard cubic feet (scf) at 5000 psig within 20 minutes of the membrane reaching operating temperature.

b. Quality Requirements:

(1) Minimum purity: 95% nitrogen gas by volume excluding other inert gases.

(2) Hydrocarbon content: Less than 50 parts per million by volume as methane.

(3) Nitrogen dryness: Atmospheric dew point of -65 F (less than 26.3 ppm water vapor by volume of gas as it enters the receiver).

(4) Particulate contamination: Less than 25 microns size using 5 or less micron filter at or close to the nitrogen gas receiver.

8. The operating temperature range of the SGSNC shall be -40°F to +130°F at altitudes up to 5000 (8) feet above sea level.

9. Nitrogen receivers.

a. Nitrogen storage capacities. High pressure gas storage of at least 600 scf at 5000 psig. Receivers shall be Department of Transportation (DOT), Type DOT 3AA. The receiver material shall permit the safe filling of receivers at temperatures of -20°F and colder.

b. Stored nitrogen: minimum of 200 scf at 5000 psig available within 20 minutes of nitrogen system engagement between the temperature of -25°F and +125°F without external systems.

c. There shall be a manual method to isolate all receivers for system operation and maintenance.

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d. A method shall be provided to prevent the entry of back pressure to the receivers in the event receiver pressure is lower than the equipment being serviced.

e. There must be a method to prevent the entry of high pressure nitrogen gas into the low pressure nitrogen gas system.

f. The receivers shall not be installed by welding. installation shall be field removable and/or replaceable.

10. Pressure Outlets. One low pressure outlet (0-350 psig) and one high pressure outlet (350-5000 psig) at or near the control panel are required for nitrogen gas. The design of the outlets shall prevent connecting the low pressure hose to the high pressure nitrogen outlet.

11. Metered port. One metered port with shutoff valve is required for connecting a portable oxygen gas analyzer. The metered port must be located on the maintenance panel. The metered port shall be compatible with the portable oxygen gas analyzer selected for use.

12. Air separation module.

a. The air separation module shall be a hollow fiber membrane which can be functionally integrated with either oil-lubricated, oil-free, and/or oil-less type compressors.

b. The membrane shall incorporate the necessary piping/plumbing and controls to isolate and by-pass the membrane and filter by connection of a high pressure line or hose.

c. Coalescent filters shall be installed on the air feed line to the membrane (to prevent contamination of the membrane) and immediately before the receivers. Oil free and oil-less type compressors are exempt from meeting this salient characteristic.

13. Operation in tilted position. The SGNSC shall function satisfactorily in a level position (horizontal oosition) and while its plane of operation is 15 degrees from horizontal in any direction.

14. Engine.

a. The engine shall be multi-fuel capable with JP-8 as the primary fuel.
The SGNSC engine shall be compatible with fuels JP-5; DF-1; DF-2; JET A; and JET A1 for start and operation.

b Fuel tank capacity shall be sufficient to permit operation of the engine for 5 hours continuous operation under load without refueling using JP-8.

c. The engine shall be capable of unaided start and operation at 24 hour soaked temperatures from -25 F to +125 F. Three unaided starts at-soaked temperatures down to -25 F.

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are required. The operational range of the engine shall be -40 F to +130 F. at altitudes from sea level to 5000 feet above sea level.

15. System Safety.

a. The SGNSC shall incorporate a safety feature that vents all system and receiver pressure above 6000 psig.

b. The contractor shall perform the following System Safety Tasks IAW MIL-STD-882C.

TASK	TITLE	TAILORED APPLICATION
101	System Safety Program	100.1 thru 101.2.5
107	System Safety Progress Summary	107.1 thru 107.2.e
301	Safety Assessment	301.1 thru 301.3.c

Task 101. All of Paragraph 4, General Requirements, is applicable. Satisfactory hazard control shall be reduction to a combined seventy- probability of IE, IID, IIIC, IVB or less IAW MIL-STD-882. Hardware configuration shall meet the criteria of MIL-STD-454M, Requirement 1 and MIL-STD-1472D, Para 5.4, 5.9 and 5.13. Identified hazards related to electrical shock shall specify the frequency, current and voltage ratings of the exposure. As a minimum. the fault hazard analysis technique or equivalent shall be used to develop system safety data.

Task 107. The analysis shall summarize open and newly-identified hazards, modifications, safety-related engineering change proposals, and changes in risks of hazards. Program activities that impact system safety shall also be evaluated (DU-SAFT-80105A/T).

Task 301, The Contractor shall summarize hazards and corrective actions pertaining to design safety features, energy sources, transportation, operations, material handling and maintenance. Use of a hazard risk index after each corrective action is required (DI-SAFT-80102A/T). Technical Orders. The Contractor shall ensure that Air Force Occupational Safety & Health standards and other applicable safety standards are addressed in all publications. All manuals shall comply with the safety requirements in MIL-M-38784C.

16. Automatic Unloader.

a. The SGNSC shall provide for high pressure nitrogen gas output at a range from 350 to 5000 psig and low pressure nitrogen gas output at a range from 0 to 350 psig with low and high pressure output hoses arid connections at or near the operators panel.

b. The SGNSC shall be fully automatic once the nitrogen system is engaged by the operator and shall remain automatic until the nitrogen system is disengaged by the operator.

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c. The SGNSC shall automatically unload when the receiver pressure reaches a set range between 3000 and 5000 +/- 200 psig and automatically load when receiver pressure drops 500 +/- 100 psig below the unload set pressure. The unload set pressure shall be adjustable between 3000 and 5000 psig.

d. The compressor shall be automatically unloaded when stopped and shall remain unloaded upon start up. The compressor shall be rated for 5000 psig service.

e. Nitrogen receiver pressure shall not be lost during compressor unloading or unit shutdown.

f. The inter-after cooler separators shall automatically dispose of any condensed moisture. oil, and/or other foreign matter but not directly into the environment, except for oil-free moisture.

17. Corrosion, Finishes and Color.

a. The SGNSC shall be corrosion resistant by the use of the best commercial practice corrosion prevention techniques and materials.

b. Finishes (primer and topcoat) shall be environmentally compliant materials producing low volatiles of carbon (VOC).

c. Color shall be in accordance with FED-STD-595, color number 24052.

18. General/Miscellaneous,

a. Natural and/or synthetic rubber shall not be painted, greased, nor oiled.

b. Threads of adjustable parts. such as tie rods, turnbuckles. etc., shall be lubricated and protected before and after assembly to prevent seizing and/or corrosion.

REGULATORY REQUIREMENTS.

19. Prohibited Materials.

a. The use of ozone depleting compounds (ODCs)/substances and the seventeen hazardous materials listed in the EPA-17 list, are all prohibited from use in any operation. maintenance, repair, and/or troubleshoot action required of the government by any SGNSC technical manual of any kind, whether it be a commercial or government format. (The EPA-17 list is published by the Environmental Protection Agency.)

b. No technical manual shall contain any of the prohibited materials for either use or information. No technical manual shall require the use of any material. cleaner, primer, or topcoat which produces high volatiles of carbon (VOC).

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c. The use of JP-4 as a test fuel is prohibited and shall not be used.

d. The use of ether as a starting aid is prohibited.

20. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practical.

BID SAMPLE TEST REQUIREMENTS.

21. Bid Sample Test Requirements. The offeror shall accomplish a bid sample test which meets the requirements of Appendix A to this CID. The bid sample test report shall meet the requirements of Appendix B to this CID and shall be delivered with the proposal.

CERTIFICATION REQUIREMENTS

22. Quality Assurance. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description and that the product conforms to the producer's own drawings, specifications, standards, arid quality assurance practices. The certificate shall be delivered with the proposal. The certificate shall meet the requirements of Appendix C to this CID.

23. Configuration Control Requirements. The contractor shall certify arid maintain substantiating evidence that all production units to be provided to the government under the contract shall be identical to the SGNSC configuration that successfully passed the bid sample test. Accordingly, to assure this configuration control requirement is met, the offeror shall comply with the following:

a. Establish the SGNSC baseline configuration to that of the bid sample which has passed all bid sample testing requirements.

b. Establish a specific model number and/or serial numbering scheme for the production units that will enable identification and control oft he production configuration to that of the bid sample.

c. Maintain engineering release records for subsequent modifications to the design of the SGNSC to assure control of the production configuration.

24. Market Acceptability Criteria.

a. The offeror shall provide data with the proposal to substantiate the sale or licensing to sell to the US Government, to a foreign government(s) and/or to the general public a specific SGNSC model which either meets the salient features of this CID, or could meet the salient features with minor modification. The following are required with the proposal:

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(1) Copies of customer purchase reequest(s) or customer signed delivery receipts to validate ten or more units sold within the last three years.

(2) Type and length of warranty offered with the sale of the above units (parts, defects, latent, or other).

(3) List of the major users of the units sold, including points of contact

(4) Forecast of the time that parts and service will be available for the units sold.

b. If the offeror has not sold tenor more SGNSC units, the following shall be provided with the proposal to substantiate that a SGN SC meeting the salient features of the CID, or a unit that could meet the features with minor modification, has been planned and will be available for sale within 120 days of the date of award. The following are required with the proposal to substantiate such planning:

(1) A production plan indicating a company-coordinated strategy for production of a specific SGNSC model. The plan shall indicate subcontractors, suppliers, and vendors of the SGNSC components/parts, quality provisions, and a time-phased manufacturing/system assembly process to demonstrate that a SGNSC shall be available 120 days after contract award.

(2) A marketing plan specifying marketing strategy, projected sales, and targeted customers.

(3) A thorough description of warranty provisions.

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APPENDIX A

SELF GENERATING NITROGEN SERVICING CART (SGNSC)

BID SAMPLE TEST REQUIREMENTS

BID SAMPLE TEST REQUIREMENTS.

1. The bid sample is the SGNSC unit which the offeror will use for bid sample testing. The bid sample shall bean operational, production-configured unit.

2. Test Facility. The test facility shall bean independent facility/laboratory which meets the following requirements:

a. For low temperature testing, the test facility shall meet the requirements of MIL-STD-810E, Method 502.3, Section II, Paragraph II-1.1 (Test Facility) and Paragraph II-1.2 (Controls).

b. For high temperature testing. the test facility shall meet the requirements of MIL-STD-810E, Method 501.3, Section II, Paragraph II-1.1 (Test Facility) arid Paragraph II-1.2 (Controls).

c. For both cold and high temperature testing, the facility shall enable the bid sample SGNSC engine and compressor to ingest conditioned air during each three hour test at each constant test temperature. (Temperature cycling during testings not required).

3. Bid Sample Test:

a. Cold temperature operation. The SGNSC shall be operated and tested in accordance with MIL-STD-810E, Method 502.3, Section I; Section II-3.2 (Procedure II-Operation); and Section II-4. The test temperatures shall be 0 F.. -10 F.. and -25 F., at ambient humidity, which shall be recorded. A twenty-four (24) hour soak is required at each test temperature. No external starting aids maybe used that are not contained within the SGNSC. Ether shall not be used as a starting aid. The engine and the compressor shall both ingest conditioned air during testing, While under test, the SGNSC shall meet the nitrogen gas production and quality requirements of' par. 7. of the CID for three consecutive how-sat each test temperature. The test fuel shall be JP-8. After the 3-hour test and at each test temperature, the SGNSC shall demonstrate the capability to have 200 scf of nitrogen gas at 5000 psig available within 20 minutes of' the nitrogen system engagement without external systems. Time to fill all high pressure receivers to a minimum total of 600 scf at 5000 psig shall not exceed 40 minutes. The receivers shall be empty at the start of each test at each test temperature (vent to atmosphere).

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b. High temperature operation. The SGNSC shall be operated and tested in accordance with MIL-STD-81OE, Method 501.3, Section I; Section II-3.2 (Procedure II- Operation); and Section II-4. The test temperatures shall be +80°F at 90% relative humidity (rh); +11O F at 80% rh and +125 F at 50% rh. .4 twenty-four (24) hour soak is required at each test temperature. The engine and the compressor shall both ingest conditioned air during testing. While under test, the SGNSC shall meet the nitrogen gas production and quality requirements of para 7. of the CID for three (3) consecutive hours at each of these temperature/humidity test levels. The test fuel shall be JP-8. After the 3 hour test and at each test temperature, the SGNSC shall demonstrate the capability to have 200 scf of nitrogen gas at 5000 psig available within twenty (20) minutes of the nitrogen gas system engagement. Time to fill all high pressure receivers to a minimum of 600 scf at 5000 psig shall not exceed 40 minutes. The receivers shall be empty at the start of each test at each test temperature (vent to atmosphere).

c. Operation in a tilt position. The SGNSC shall meet the nitrogen gas production and quality requirements of paragraph 7. of the CID for one (1) hour when tilted at 15 degrees in any direction from the level position while at ambient altitude, temperature, and humidity. The test fuel shall be JP-8. After the one hour test. and while still tilted 15 degrees, nitrogen gas production shall be sufficient to have 200 scf of nitrogen gas at 5000 psig available within twenty minutes of nitrogen gas system engagement. The time to fill all high pressure receivers to a minimum 600 scf at 5000 psig shall not exceed forty (40) minutes. The receivers shall be empty at the start of each test (vent to atmosphere).

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APPENDIX B

SELF GENERATING NITROGEN SERVICING CART (SGNSC)

TEST REPORT REQUIREMENTS

1. The bid sample test report shall contain the following minimum requirements:

a. Description of bid sample, including specific manufacturer, model/serial/part number, as applicable, and photograph.

b. Description of test procedure, test duration, and test conditions.

c. Number of trials for which a specific measurement is made and/or a given test procedure is accomplished.

d. Description of how the bid sample was operated during test.

e. Number of thermocouples and their specific location on the bid sample. Minimum of four thermocouples required.

f. Test set-up diagram/photos. Show arrangement of bid sample relative to test equipment used.

g. List of test equipment used during test. Identify model, manufacturer, and calibration status of' all test equipment used.

h. Test results at each test temperature condition:

(1) Nitrogen gas production rate in scf/minute. Determine and document rate every thirty minutes during each three hour test.

(2) Nitrogen purity excluding all other inert gases. Determine and document every thirty minutes during each three hour test.

(3) Determine and document atmospheric dew point every thirty minutes during each three hour test.

(4) Determine and document methane content in parts per million by gas chromatography every thirty minutes during each three hour test.

(5) Time for the membrane to reach operating temperature at which the required nitrogen purity can be produced during testing at temperatures below 100°F.

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(6) Thermocouple readings at end of twenty four hour soak cycle.

(7) Number of cold start attempts and results.

(8) Time to produce 200 scf of 95% pure nitrogen gas at 5000 psig after nitrogen system engagement.

(9) Time to fill all high pressure receivers to 600 scf at 5000 psig.

(10) Data analysis results.

2. The report shall be delivered to the government with the offeror's proposal.

3. The report shall be signed by both an officer of the test laboratory and by an officer of the offeror who has the authority to bind the company contractually.

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APPENDIX C

SELF GENERATING NITROGEN SERVICING CART (SGNSC)

CERTIFICATE FORMATS

The required certifications shall bear the signature of a company official with the authority to bind the company contractually.

QUALITY ASSURANCE

(Para 22 of the CID)

I hereby certify that (Company's Name) has maintained substantiating evidence that the SGNSC offered meets all of the salient characteristics of the Commercial item Description and that the product conforms to

(Company's Name) drawings, specifications, standards and quality assurance practices.

I understand the government reserves the right to require proof of such conformance prior to first delivery and thereafter.

CONFIGURATION CONTROL REQUIREMENTS

(Para 23 of the CID)

I hereby certify that (Company's Name) has maintained substantiating evidence that all production units to-be provided to the government under the contract shall be identical to the SGNSC configuration that successfully passed the bid sample test. I further certify that (Company's Name)

a. Has established a SGNSC baseline configuration to that of the successful bid sample.

b. Has established a specific model number and/or serial numbering scheme fob the production units that will enable identification and control of the production configuration to that of the bid sample.

c. Will maintain engineering release records for subsequent modification to the design of the SGNSC to assure control of the production configuration.

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MARKET ACCEPTABILITY CRITERIA CERTIFICATE

(Para 24 of the CID)

"X" Applicable and attach appropriate information.

_____ (Company's Name) has provided the information prescribed in paragraph 24.a of the SGNSC CID to substantiate the sale or licensing to sell the U.S. Government, to a foreign government(s) and/or to the general public a specific SGNSC model which either meets the salient features on the CID, or could meet the salient features with minor modification. IAW Para 24a. (1), (2), (3) and (4) of the CID.

_____ (Company's Name) has provided with the proposal the information prescribed in paragraph 24.b of the SGNSC CID to substantiate that a SGNSC meeting the salient features of the CID, or a unit that could meet the features with minor modification has been planned and will be available for sale within 120 days of the date of the contract award. The following are provided with the proposal to substantiate such planning:

1. A production plan IAW CID, para 24.b.(1).

2. A marketing plan IAW CID, para 24.b.(2).

3. A thorough description of warranty provisions, para 24.b.(3) of the CID.

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